Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A student appears for tests I, II and III. The student is successful if he passes either in tests I and II or tests I and III. The probabilities of the student passing in tests I, II, III are p, q and $\frac{1}{2}$ respectively. The probability that the student is successful is $\frac{1}{2}$ then the relation between p and q is given by

(A) $\mathrm{pq}+\mathrm{p}=1$

(B) $\mathrm{p}^2+\mathrm{q}=1$

(C) $\mathrm{pq}-1=\mathrm{p}$

(D) None of these

Q.2

A bag contains 4 red and 4 blue balls. Four balls are drawn one by one from the bag, then find the probability that the drawn balls are in alternate colour.

(A) $\frac{35}{6}$

(B) $\frac{2}{35}$

(C) $\frac{3}{35}$

(D) $\frac{6}{35}$

Q.3

In an examination, the probability of a candidate
solving a question is $\frac{1}{2}$. Out of given 5 questions in the examination, what is the probability that the candidate was able to solve at least 2 questions?

(A) $\frac{1}{64}$

(B) $\frac{3}{16}$

(C) $\frac{1}{2}$

(D) $\frac{13}{16}$

Q.4

It has been found that if A and B play a game 12 times, A wins 6 times, B wins 4 times and they draw twice. A and B take part in a series of 3 games. The probability that they win alternately, is:

(A) $5/12$

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(B) $5/36$

(C) $19/27$

(D) $5/27$

Q.5

An experiment consists of flipping a coin and then flipping it a second time if head occurs. If a tail occurs on the first flip, then a six-faced die is rolled once. Assuming that the outcomes are equally likely, what is the probability of getting one head and one tail?

(A) $1/4$

(B) $1/36$

(C) $1/6$

(D) $1/8$

Q.6

Rajesh doesn't like to study. Probability that he will study is $1/3$. If he studied then probability that he will fail is $1/2$ and ifhe didn't study then probability that he will fail is $3/4$. If in result Rajesh failed then what is the probability that he didn't studied.

(A) $2/3$

(B) $3/4$

(C) $1/3$

(D) None of these

Q.7

Two aeroplanes I and II bomb a target in suecession. The probabilies of I and II scoring a hit correctly are 0.3 and 0.2 respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane is

(A) 0.2

(B) 0.7

(C) 0.06

(D) 0.14

Q.8

A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is

(A) $\frac{8}{729}$

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(B) $\frac{8}{243}$

(C) $\frac{1}{729}$

(D) $\frac{8}{9}$

Q.9

For $k=1,2,3$ the box B_k contains k red balls and $(k+1)$ white balls. Let $\mathrm{P}\left(\mathrm{B}_1\right)=\frac{1}{2}, \mathrm{P}\left(\mathrm{B}_2\right)=\frac{1}{3}$ and $\mathrm{P}\left(\mathrm{B}_3\right)=\frac{1}{6}$. A box
is selected at random and a ball is drawn from it. If a red ball is
drawn, then the probability that it has come from box B_2, is

(A) $\frac{35}{78}$

(B) $\frac{14}{39}$

(C) $\frac{10}{13}$

(D) $\frac{12}{13}$

Q.10

A bag contains n balls. It is given that the probability that among these n balls exactly r balls are white is proportional to $r^2(0 \leq r \leq n)$. A ball is drawn at random and is found to be white. Then the probability that all the balls in the bag are white, will be:

(A) $\frac{2n}{(n+1)^2}$

(B) $\frac{4n}{(n+1)^2}$

(C) $\frac{2n}{(n+3)^2}$

(D) $\frac{4n}{(n+3)^2}$

Q.11

In a test, an examinee either guesses or copies or knows the answer to a multiple choice question with four choices. The probability that he makes a guess is $\frac{1}{3}$. The probability that he copies is $\frac{1}{6}$ and the probability that his answer is correct
given that he copied it is $\frac{1}{8}$. The probability that he knew the answer to the question given that he correctly answered it, is

(A) $\frac{24}{29}$

(B) $\frac{1}{4}$

(C) $\frac{3}{4}$

(D) $\frac{1}{2}$

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Q.12

A and B are two independent witnesses (i.e. there is no collision between them) in a case. The probability that A will speak the truth is x and the probability that B will speak the truth is y. A and B agree in a certain statement. The probability that the statement is true is

(A) $\frac{x-y}{x+y}$

(B) $\frac{xy}{1+x+y+xy}$

(C) $\frac{x-y}{1-x-y+2xy}$

(D) $\frac{xy}{1-x-y+2xy}$

Q.13

A card from a pack of 52 cards is lost. From the remaining cards of the pack two cards are drawn and found to be hearts. If the probability that lost card is hearts, is k, then the value of $100 \quad k$ is

(A) 22

(B) 33

(C) 44

(D) 55

Q.14

There are four machines and it is known that exactly two of them are faulty. They are tested, one by one, in a random order till both the faulty machines are identified. Then the probability that only two tests are needed is

(A) $\frac{1}{3}$

(B) $\frac{1}{6}$

(C) $\frac{1}{2}$

(D) $\frac{1}{4}$

Q.15

An urn contains 6 white and 4 black balls. A fair die is rolled and that number of balls are chosen from the urn. Find the probability that balls selected are white.

(A) 0.3

(B) 0.9

(C) 0.2

(D) 0.5

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Q.1 Answer:

$pq + p = 1$

Solution:

Let A, B and C be the events that the student is successful in tests I, II and III respectively. Then P**(The student is successful)** $= P(A)P(B)\{1 - P(C)\} + P(A)\{1 - P(B)\}P(C)$
$+P(A)P(B)P(C)$
$= p \cdot q\left(1 - \frac{1}{2}\right) + p(1 - q)\frac{1}{2} + p \cdot q\frac{1}{2}$
$= \frac{1}{2}pq + \frac{1}{2}p(1 - q) + \frac{1}{2}pq$
$= \frac{1}{2}(pq + p - pq + pq) = \frac{1}{2}(pq + p)$
$\therefore \frac{1}{2} = \frac{1}{2}(pq + p) \Rightarrow 1 = pq + p$

Q.2 Answer:

$\frac{6}{35}$

Solution:

E_1 **: Event that first drawn ball is red, second is blue and so on.** E_2 **: Event that first drawn ball is blue, second is red and so on.** $\therefore P(E_1) = \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5}$ **and** $P(E_2) = \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5}$
$P(E) = P(E_1) + P(E_2) = 2 \times \frac{4}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{3}{5} = \frac{6}{35}$

Q.3 Answer:

$\frac{13}{16}$

Solution:

$$P = {}^5C_2\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^3 + {}^5C_3\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 + {}^5C_4\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^1 + {}^5C_5\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^0$$
$$= \left(\frac{1}{2}\right)^5\left[{}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5\right]$$
$$= \frac{1}{2^5}[10 + 10 + 5 + 1]$$
$$= \frac{1}{2^5} \times 26 = \frac{13}{16}$$

Q.4 Answer:

$5/36$

Solution:

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$P(A) = \frac{6}{12} = \frac{1}{2}, P(B) = \frac{4}{12} = \frac{1}{3}$

Req. probability

$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{12} + \frac{1}{18} = \frac{5}{36}$

Q.5 Answer:

$1/4$

Solution:

Required probability $= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Q.6 Answer:

$3/4$

Solution:

He will fail in exam in two cases:

Case (i) He studied and failed, probability of this case is $(1/3)(1/2) = 1/6$ **Case (ii) He didn't studied and failed, probability of this case is** $(2/3)(3/4) = 1/2$**. So total probability is** $1/6 + 1/2 = 4/6 = 2/3$ **Then required probability** $= (1/2)/(2/3) = 3/4$**.**

Q.7 Answer:

0.14

Solution:

If we assume that only once they are hitting the target then required probability

$= (0.7)(0.2) = 0.14$

Q.8 Answer:

$\frac{8}{243}$

Solution:

Probability of getting 9 in a single throw

$= \frac{4}{36} = \frac{1}{9}$

$\therefore$ **Total probability** $= {}^3C_2 \times \left(\frac{1}{9}\right)^2 \times \frac{8}{9} = \frac{8}{243}$

Q.9 Answer:

$\frac{14}{39}$

Solution:

In a box, $B_1 = 1R, 2\ W; B_2 = 2R, 3\ W$ **and** $B_3 = 3R, 4\ W$

Also, given that, $P(B_1) = \frac{1}{2}, P(B_2) = \frac{1}{3}$ **and** $P(B_3) = \frac{1}{6}$

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$P\left(\frac{B_2}{R}\right)=\frac{P(B_2)P\left(\frac{R}{B_2}\right)}{P(B_1)P\left(\frac{R}{B_1}\right)+P(B_2)P\left(\frac{R}{B_2}\right)+P(B_3)P\left(\frac{R}{B_3}\right)}$

$=\frac{\frac{1}{3}\times\frac{2}{5}}{\frac{1}{2}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{5}+\frac{1}{6}\times\frac{3}{7}}=\frac{\frac{2}{15}}{\frac{1}{6}+\frac{2}{15}+\frac{1}{14}}=\frac{14}{39}.$

Q.10 Answer:

$\frac{4n}{(n+1)^2}$

Solution:

Let $E_i=0,1,2,\ldots n$ be the event that the bag contains exactly i white balls then $p\left(E_i\right)k^2i$ where $\sum_{i=0}^{n}k_i^2=1\Rightarrow k=\frac{r}{n(n+1)(2n+1)}$

write this as new solution

Let A be the event that a ball drawn is white

$p\left(\frac{E_n}{A}\right)=\frac{p(E_n)p(A/E_n)}{\sum_{i=0}^{n}k_i^2\left(\frac{i}{n}\right)}=\frac{K\cdot n^2\cdot 1}{\frac{k}{n}\sum_{t=0}^{n}(i)^3}=\frac{n^3}{\left(\frac{n(n+1)}{2}\right)^2}$

$p\left(\frac{E_n}{A}\right)=\frac{4n}{(n+1)^2}$

Q.11 Answer:

$\frac{24}{29}$

Solution:

Let E_1 be the event that the answer is guessed, E_2 be the event that the answer is copied, E_3 be the event that the examinee knows the answer and ${}^3\mathrm{E}$ be the event that the examinee answers correctly.

Given $\mathrm{P}\left(\mathrm{E}_1\right)=\frac{1}{3},\mathrm{P}\left(\mathrm{E}_2\right)=\frac{1}{6}$,

Assume that events $E_1,E_2\&E_3$ are exhaustive. $\therefore\mathrm{P}\left(\mathrm{E}_1\right)+\mathrm{P}\left(\mathrm{E}_2\right)+\mathrm{P}\left(\mathrm{E}_3\right)=1$

$\therefore\mathrm{P}\left(\mathrm{E}_3\right)=1-\mathrm{P}\left(\mathrm{E}_1\right)-\mathrm{P}\left(\mathrm{E}_2\right)=1-\frac{1}{3}-\frac{1}{6}=\frac{1}{2}.$

Now, $\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{E}_1}\right)\equiv$ Probability of getting correct answer

by guessing $=\frac{1}{4}$ (Since 4 alternatives) $\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{E}_2}\right)\equiv$ Probability of answering correctly by copying $=\frac{1}{8}$ and $\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{E}_3}\right)\equiv$ Probability of answering correctly by knowing $=1$

Clearly, $\left(\frac{\mathrm{E}_3}{\mathrm{E}}\right)$ is the event he knew the answer to the question given that he correctly answered it.

Using Baye's theorem $\mathrm{P}\left(\frac{\mathrm{E}_3}{\mathrm{E}}\right)$

$=\frac{\mathrm{P}(\mathrm{E}_3)\cdot\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{E}_3}\right)}{\mathrm{P}(\mathrm{E}_1)\cdot\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{E}_1}\right)+\mathrm{P}(\mathrm{E}_2)\cdot\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{E}_2}\right)+\mathrm{P}(\mathrm{E}_3)\cdot\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{E}_3}\right)}$

$=\frac{\frac{1}{2}\times1}{\frac{1}{3}\times\frac{1}{4}+\frac{1}{6}\times\frac{1}{8}+\frac{1}{2}\times1}=\frac{24}{29}$

Q.12 Answer:

$\frac{xy}{1-x-y+2xy}$

Solution:

A and B will agree in a certain statement if both speak truth or both tell a lie. We define following events $\mathrm{E}_1=\mathrm{A}$ and B both speak truth $\Rightarrow\mathrm{P}\left(\mathrm{E}_1\right)=\mathrm{xy}$

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$\mathrm{E}_2 = \mathrm{A}$ **and** B **both tell a lie** $\Rightarrow \quad P(E_2) = (1-x)(1-y)$

$\mathrm{E} = \mathrm{A}$ **and** B **agree in a certain statement Clearly,** $\mathrm{P}(\mathrm{E}/\mathrm{E}_1) = 1$ **and** $\mathrm{P}(\mathrm{E}/\mathrm{E}_2) = 1$

The required probability is $\mathrm{P}(\mathrm{E}_1/\mathrm{E})$**. Using Baye's theorem**

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1)+P(E_2)P(E/E_2)}$$

$$= \frac{xy \cdot 1}{xy \cdot 1+(1-x)(1-y)\cdot 1} = \frac{xy}{1-x-y+2xy}$$

Q.13 Answer:

22

Solution:

E_1: **lost card is a card of hearts**

E_2: **lost card is a card of clubs**

E_3: **lost card is a card of spades**

E_4: **lost card is a card of diamonds**

A: **Drawing two cards of hearts**

$$PE_1 = \frac{1}{4} = PE_2 = PE_3 = PE_4$$

$$PA \mid E_1 = \frac{{}^{12}C_2}{{}^{51}C_2}$$

$$PA \mid E_2 = \frac{{}^{13}C_2}{{}^{51}C_2} = PA \mid E_3 = PA \mid E_4$$

Required probability $= \mathrm{PE}_1 \mid \mathrm{A}$ **where**

$$\mathrm{PE}_1 \mid \mathrm{A} = \frac{\mathrm{PE}_1 \mathrm{PA} \mid \mathrm{E}_1}{\sum_{i=1}^{4} \mathrm{PE}_i \mathrm{PA} \mid \mathrm{E}_i}$$

$$= \frac{\frac{1}{4} \cdot \frac{{}^{12}C_2}{{}^{51}C_2}}{\frac{1}{4}\frac{{}^{12}C_2}{{}^{51}C_2} + \frac{3\,{}^{13}C_2}{{}^{51}C_2}}$$

$$= \frac{66}{66+3(78)}$$

$$= \frac{11}{50} = k$$

$$\Rightarrow 100k = 22$$

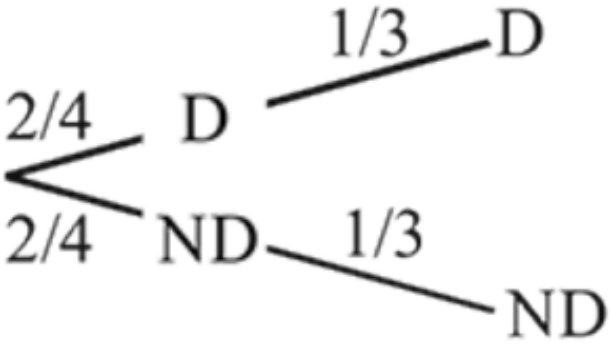
Q.14 Answer:

$\frac{1}{3}$

Solution:

The faulty machines can be identified in two tests only if both the tested machines are either all defective or all non-defective. See the following tree diagram.

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(Here D is for Defective & ND is for NonDefective) Reqd. Probability $= \frac{2}{4} \times \frac{1}{3} + \frac{2}{4} \times \frac{1}{3} = \frac{1}{3}$

$\because$ The probability that first machine is defective (or non-defective) is $\frac{2}{4}$ and the probability that second machine is also defective (or non defective) is $\frac{1}{3}$ as 1 defective machine remains in total three machines.

Q.15 Answer:

0.2

Solution:

A_i: **Event that number i appears on the die.**

E: **Event that only white balls are drawn.**

$PA_1 = PA_2 = \ \ldots \ = PA_6 = \frac{1}{6}$

$PE \mid A_i = \frac{{}^{6}C_i}{{}^{10}C_i}, \quad i = 1, \quad 2, \ \ldots, \quad 6$

$\Rightarrow$ **Required Probability** $= PE$

$= P \bigcup_{i=1}^{6} E \cap A_i$

$= \sum_{i=1}^{6} PE \cap A_i$

$= \sum_{i=1}^{6} PA_i PE \mid A_i$

$= \frac{1}{6}\frac{6}{10} + \frac{15}{45} + \frac{20}{120} + \frac{15}{210} + \frac{6}{252} + \frac{1}{210}$

$= \frac{1}{6}\frac{3}{5} + \frac{1}{3} + \frac{1}{6} + \frac{1}{14} + \frac{1}{42} + \frac{1}{210}$

$= \frac{1}{6}\frac{33}{30} + \frac{4}{42} + \frac{1}{210}$

$= \frac{1}{6}\frac{11}{10} + \frac{2}{21} + \frac{1}{210}$

$= \frac{1}{6}\frac{(11)(21) + 20 + 1}{210}$

$= \frac{1}{6}\frac{(21)(12)}{210} = 0.2$

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